Data Structures

HW SET - 8

Question 10: Separate Chaining in a Hash Table

Question: Implement a hash table using separate chaining with initialize, insert, and search operations.

C Program:

```
#include <stdio.h>

#include <stdlib.h>

#define SIZE 7


typedef struct Node {

    int key;

    struct Node *next;

} Node;


Node *table[SIZE];


void initialize() {

    for (int i = 0; i < SIZE; i++)

        table[i] = NULL;

}


void insert(int key) {

    int i = key % SIZE;

    Node *p = malloc(sizeof(Node));

    p->key = key;

    p->next = table[i];

    table[i] = p;

}
```

```
int search(int key) {
    for (Node *p = table[key % SIZE]; p; p = p->next)
        if (p->key == key)
            return 1;
    return 0;
}

void display() {
    for (int i = 0; i < SIZE; i++) {
        printf("%d:", i);
        for (Node *p = table[i]; p; p = p->next)
            printf(" %d ->", p->key);
        puts(" NULL");
    }
}

int main() {
    initialize();
    insert(10);
    insert(17);
    insert(24);
    insert(5);
    display();

    printf("Search 17: %s\n",
           search(17) ? "Found" : "Not found");
    printf("Search 8: %s\n",
           search(8) ? "Found" : "Not found");
```

```
    return 0;
}
```

Sample Output:

```
0: 24 -> 17 -> 10 -> NULL
1: NULL
2: NULL
3: NULL
4: NULL
5: 5 -> NULL
6: NULL
Search 17: Found
Search 8: Not found
```